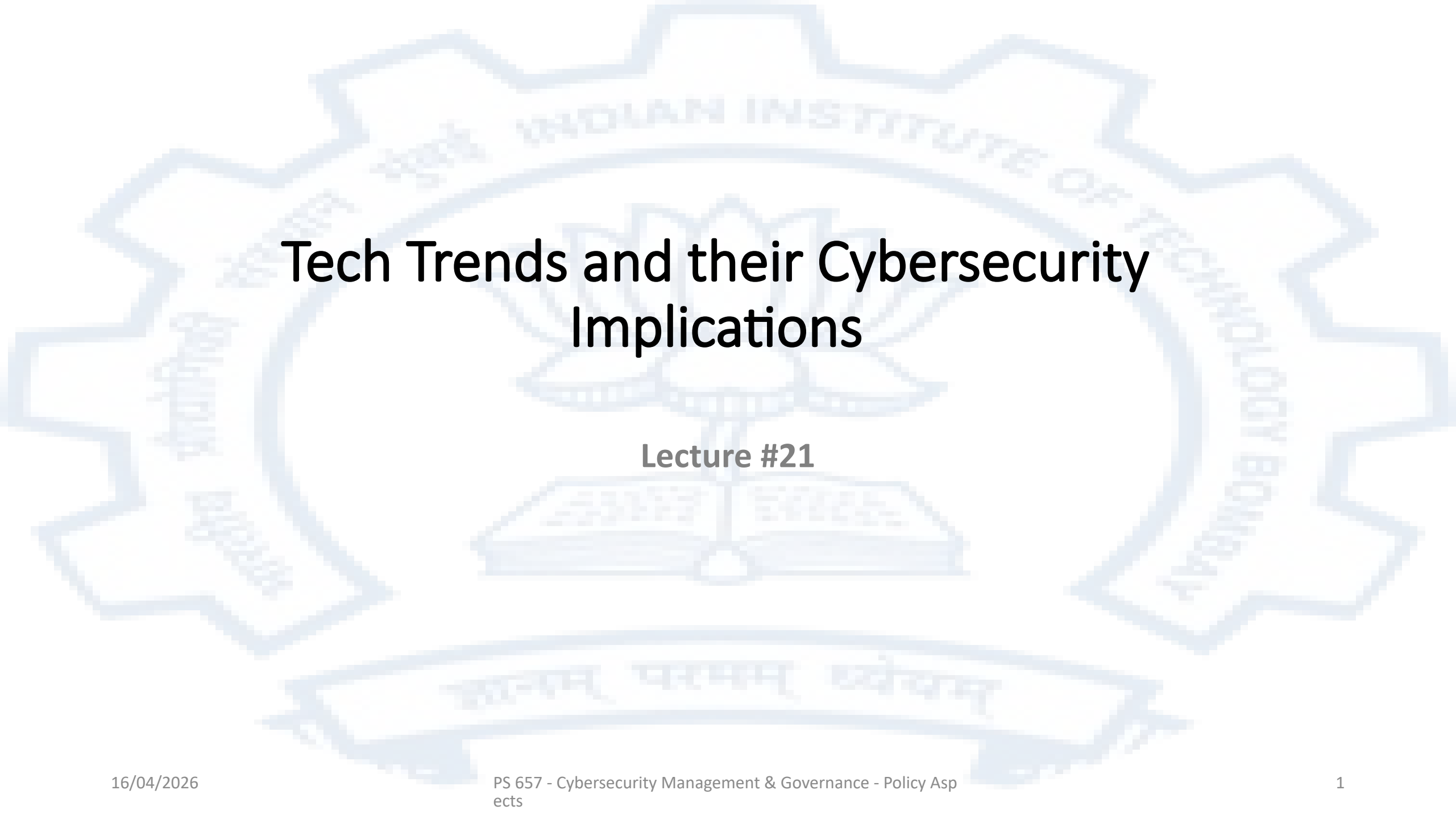




Tech Trends and their Cybersecurity Implications

Lecture #21



AI and ML

Artificial Intelligence- Definition

Artificial intelligence (AI) is intelligence demonstrated by machines, as opposed to the **natural intelligence** displayed by humans or animals. Leading AI textbooks define the field as the study of "intelligent agents": any system that perceives its environment and takes actions that maximize its chance of achieving its goals. Some popular accounts use the term "artificial intelligence" to describe machines that mimic "cognitive" functions that humans associate with the human mind, such as "learning" and "problem solving", however this definition is rejected by major AI researchers.

https://en.wikipedia.org/wiki/Artificial_intelligence

Machine Learning

Machine learning (ML) is the study of computer algorithms that can improve automatically through experience and by the use of data. It is seen as a part of artificial intelligence. Machine learning algorithms build a model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so. Machine learning algorithms are used in a wide variety of applications, such as in medicine, email filtering, speech recognition, and computer vision, where it is difficult or unfeasible to develop conventional algorithms to perform the needed tasks.

https://en.wikipedia.org/wiki/Machine_learning

Debate this.

Categories of Machine Learning

1. Supervised Learning

A system is given many samples of data that have presented as having an “inputs” part and an “outputs” part. The outputs are classified by a human expert. The learning system then attempts to find a general rule that can be used to compute the outputs from inputs even on samples that were not in the original set (“training set”).

2. Unsupervised Learning

In unsupervised learning the outputs are not classified as belonging to any category. These systems may be able to discover patterns that “experts” have not. These techniques also require more data and time to arrive at acceptable accuracy.

3. Reinforcement Learning

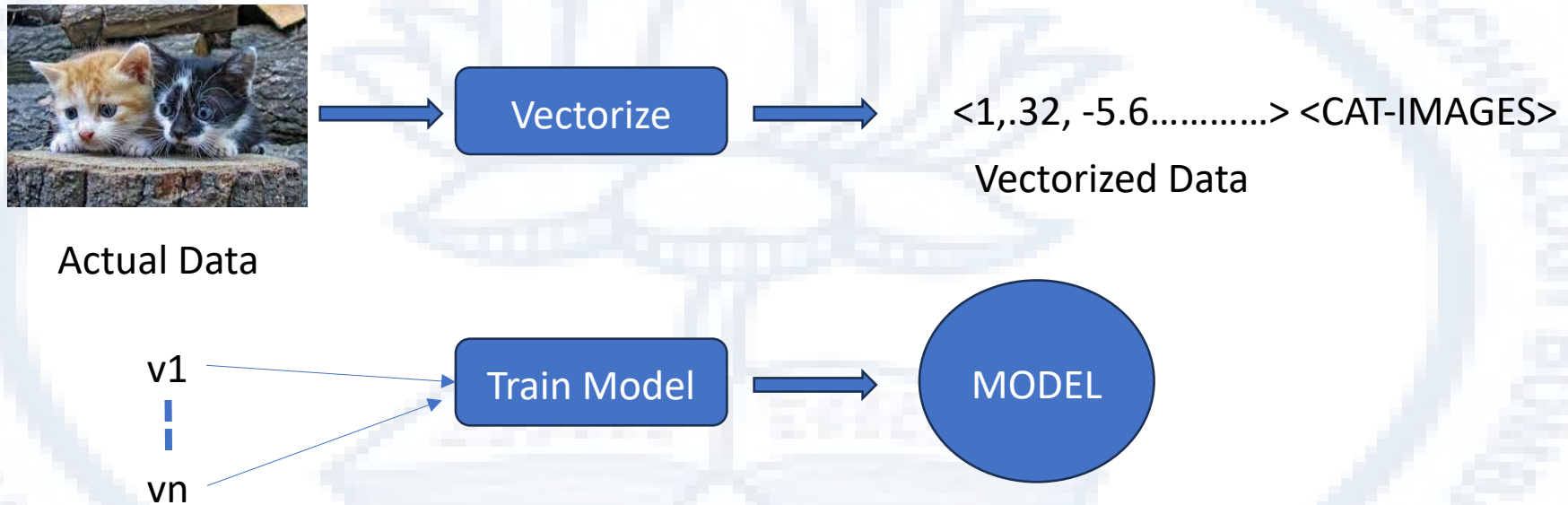
When the problem to be solved involves a series of actions that need to be taken to maximize a “reward”, the method used it to reward desirable behaviour and penalize undesirable behaviour. This paradigm is called reinforcement learning

Vectorization

- In order to use any of the machine learning algorithms, the input data must be converted into numeric values
- Each item of data, which could be an entire document, must be converted into a vector of numeric values
- This process is called vectorization
- There are some standard methods of text vectorization

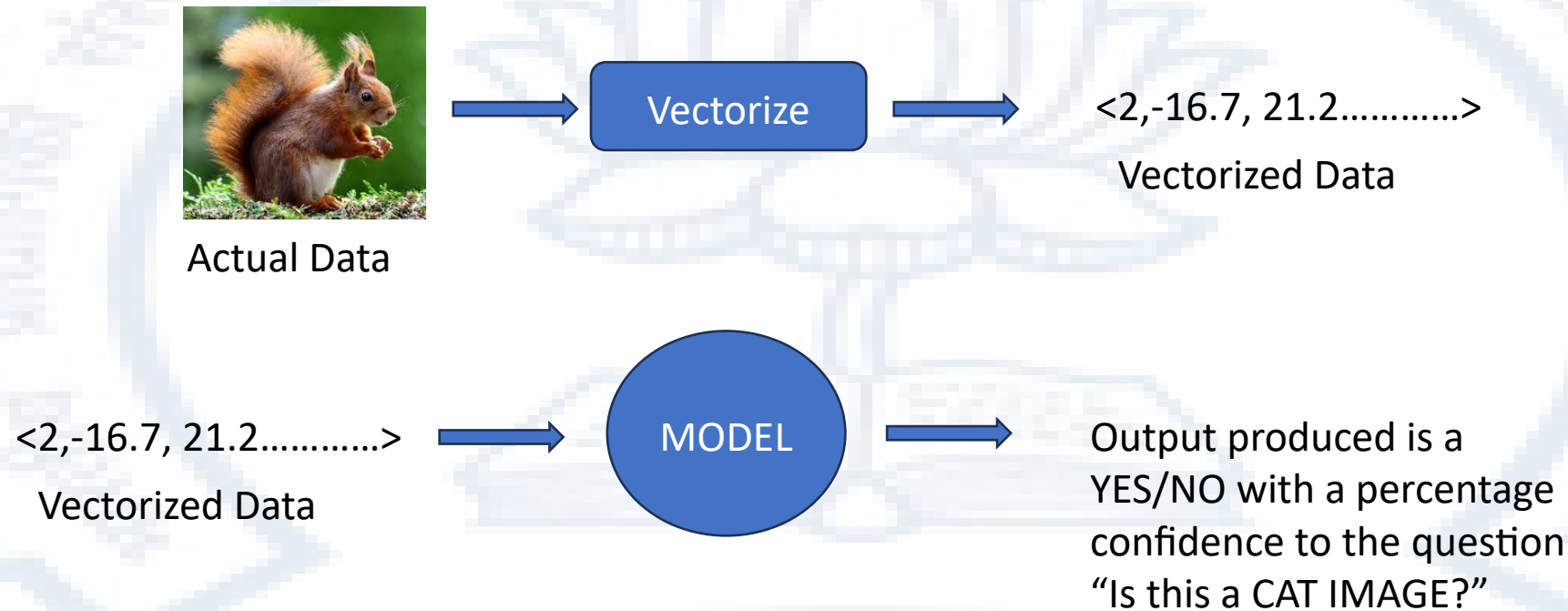
Using Unstructured Data – an example

Training the model



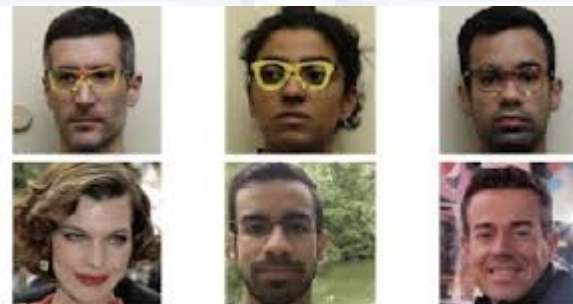
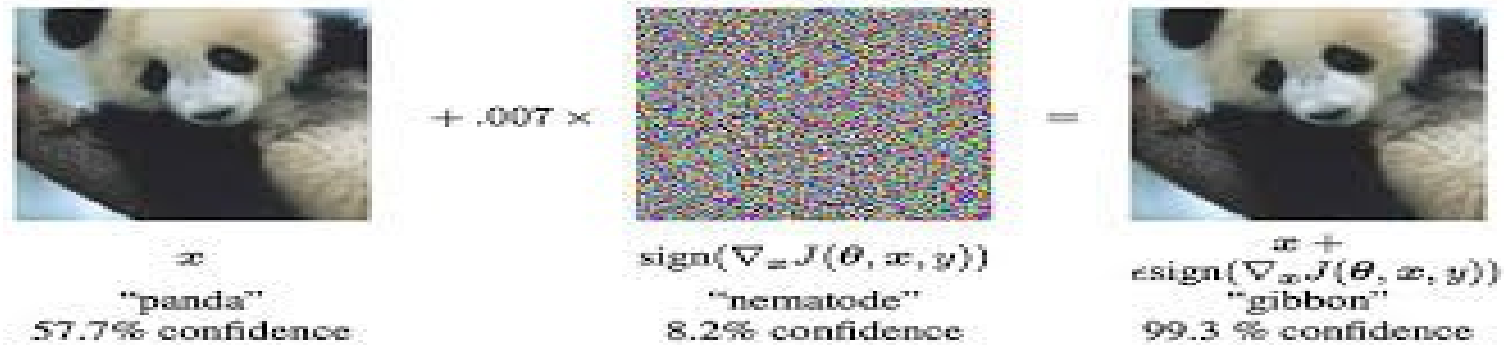
Using Unstructured Data – an example

Using the model



Risks to AI/ML Systems

- AI/ML systems are based on data and therefore security of data (CIA triad) is critical to ensuring the correct functioning of AI
 - Attacks on AI systems could be aimed at data manipulation or poisoning to make them biased or make them malfunction
 - Inadequate protection may create opportunities for model theft



Use of AI/ML systems to mount attacks

- Use of AI/ML to personalize attacks at scale. Today many phishing emails are generic but if they could be personalized at scale that may improve the success rate of the attack
- AI/ML can be used by malware to evade detection
- Generative AI can generate deceptive content and manipulate content – deepfakes
- AI/ML can scale attacks through bots that can act autonomously

Cybersecurity issues with digital currency

- If the digital currency is not based on cryptographic technologies then forgery and double spending are issues to be addressed
- If the blockchains in use are permissioned then getting unauthorized access compromises the operation
- For public blockchains the issues are
 - Destruction of wallets
 - Hacking into wallets
 - Hacking into crypto-exchanges
 - 2014 Mt. Gox hack
 - Bugs in smart contracts
 - 2016 DAO attack on Ethereum

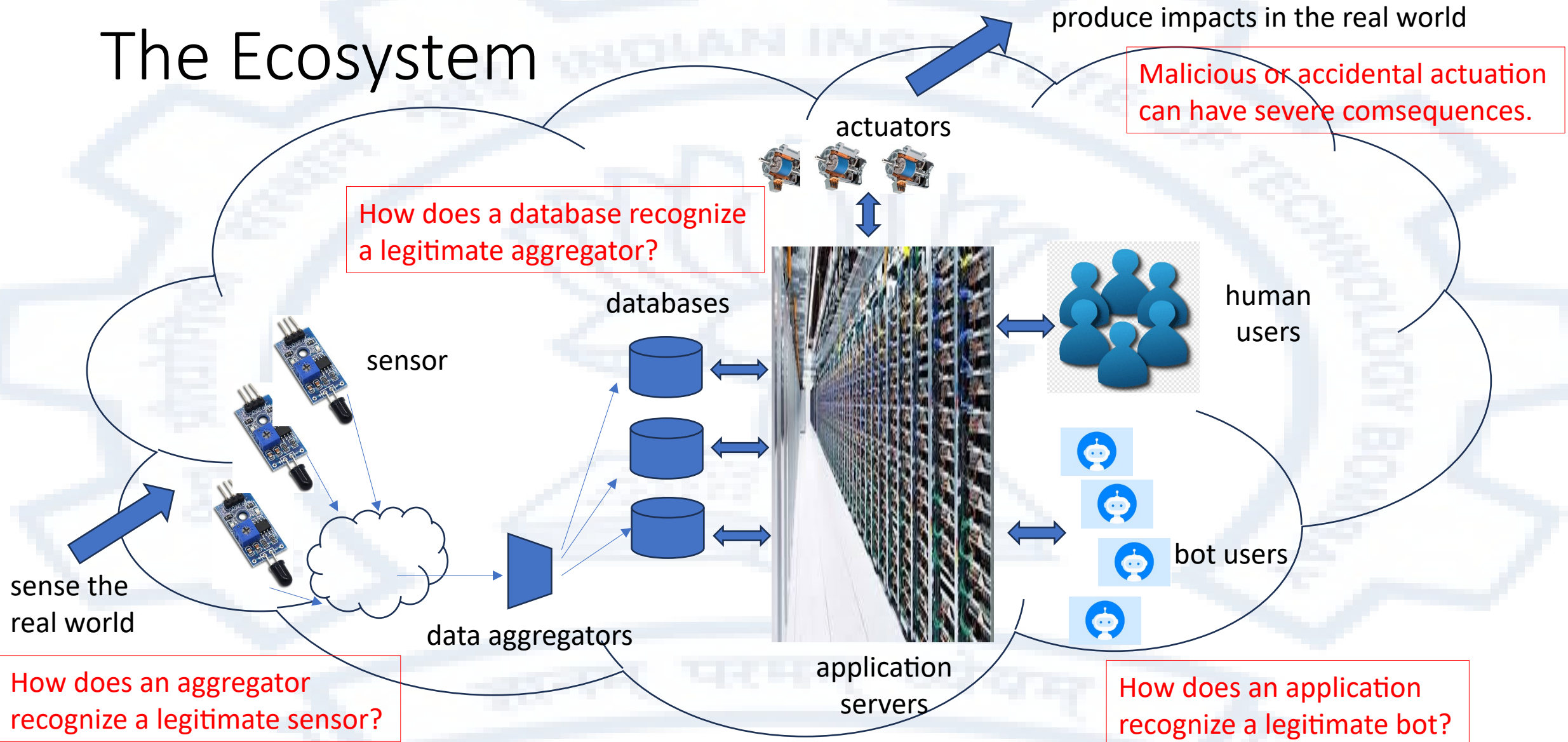
The background of the slide features a large, light blue watermark of the Indian Institute of Technology Bombay (IIT Bombay) logo. The logo is a gear-like shape with a lotus flower in the center. The text "INDIAN INSTITUTE OF TECHNOLOGY BOMBAY" is written along the top inner edge of the gear, and "सत्यम् धर्मम् विद्यया" is written along the bottom inner edge.

Other important trends

IoT and the emergence of a cyber-physical continuum – machines integrate with cyberspace

- The ability to connect “anything anywhere” to “anything anywhere”, which some call IoT, has been enabled by
 - Protocols of the Internet
 - Deployment of mobile communications networks
 - Vast increases in computing power that have made mobile phones into powerful computers
 - Availability of low-cost reliable sensors/actuators capable of transmitting and receiving data
- The reality of today is that there is not longer a clear boundary between the real world and cyber worlds. A cyber-physical continuum has emerged where events in cyberworlds can impact the real world and *vice versa*.

The Ecosystem



The Metaverse – Humans integrate with cyberspace

- *In science fiction, the **metaverse** is a hypothetical iteration of the Internet as a single, universal, and immersive virtual world that is facilitated by the use of virtual reality (VR) and augmented reality (AR) headsets.*
- *In colloquial usage, a **metaverse** is a network of 3D virtual worlds focused on social and economic connection.*
- *In scientific research, it is defined as 'a three-dimensional online environment in which users represented by avatars interact with each other in virtual spaces decoupled from the real physical world'*

Major security concerns with the Metaverse

- **Identity & Authentication Risks** Avatars and digital identities can be hijacked, leading to impersonation and fraud.
- **Data Privacy Concerns** XR devices collect biometric and behavioral data, which can be exploited if not secured.
- **Blockchain & Smart Contract Vulnerabilities** Since many metaverse economies rely on crypto-assets, flaws in blockchain protocols can lead to theft or manipulation.
- **IoT Integration Risks** Devices connected to metaverse platforms (VR headsets, sensors) expand the attack surface.
- **Social Engineering & Phishing in 3D Environments** Immersive settings make scams more convincing and harder to detect.

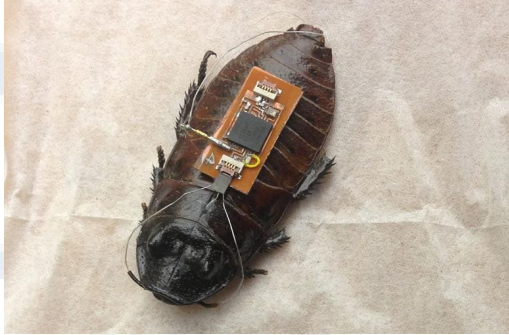
Brain Computer Interfaces – Current Status

- In November 2011 *Time* magazine selected "necomimi" produced by Neurowear as one of the best inventions of the year. The company announced that it expected to launch a consumer version of the garment, consisting of catlike ears controlled by a brain-wave reader produced by NeuroSky, in spring 2012.
- In February 2014 They Shall Walk (a nonprofit organization fixed on constructing exoskeletons, dubbed LIFESUITs, for paraplegics and quadriplegics) began a partnership with James W. Shkarji on the development of a wireless BCI.^[196]
- In 2016, a group of hobbyists developed an open-source BCI board that sends neural signals to the audio jack of a smartphone, dropping the cost of entry-level BCI to £20. Basic diagnostic software is available for Android devices, as well as a text entry app for Unity.
- In 2020, NextMind released a dev kit including an EEG headset with dry electrodes at \$399. The device can be played with some demo applications or developers can create their own use cases using the provided Software Development Kit.
- ISO has recently created a committee to develop standards in this area
- IIT Bombay has active projects in this area

ISO/IEC JTC 1/SC 43 is being supported administratively by IEC. All information related to ISO/IEC JTC 1/SC 43 is available on the [IEC web site](#)

Standardization in the area of Brain-computer Interfaces for information technology to enable communication and interaction between brain and computers that are applicable across application areas.

Bio-Digital Convergence



Scientists are already able to mount microchips and sensors onto cockroaches and other insects to control their motion by remote control and DIY kits for building your own “roboroach” are already available online.

The ability to create and manipulate biological systems using digital technology is one of the key drivers of bio-digital convergence. This includes the use of digital tools and techniques to design and synthesize new biological components, such as DNA, proteins and cells, as well as the use of digital technology to control and manipulate these biological systems in real time.

The Cybersecurity Challenges

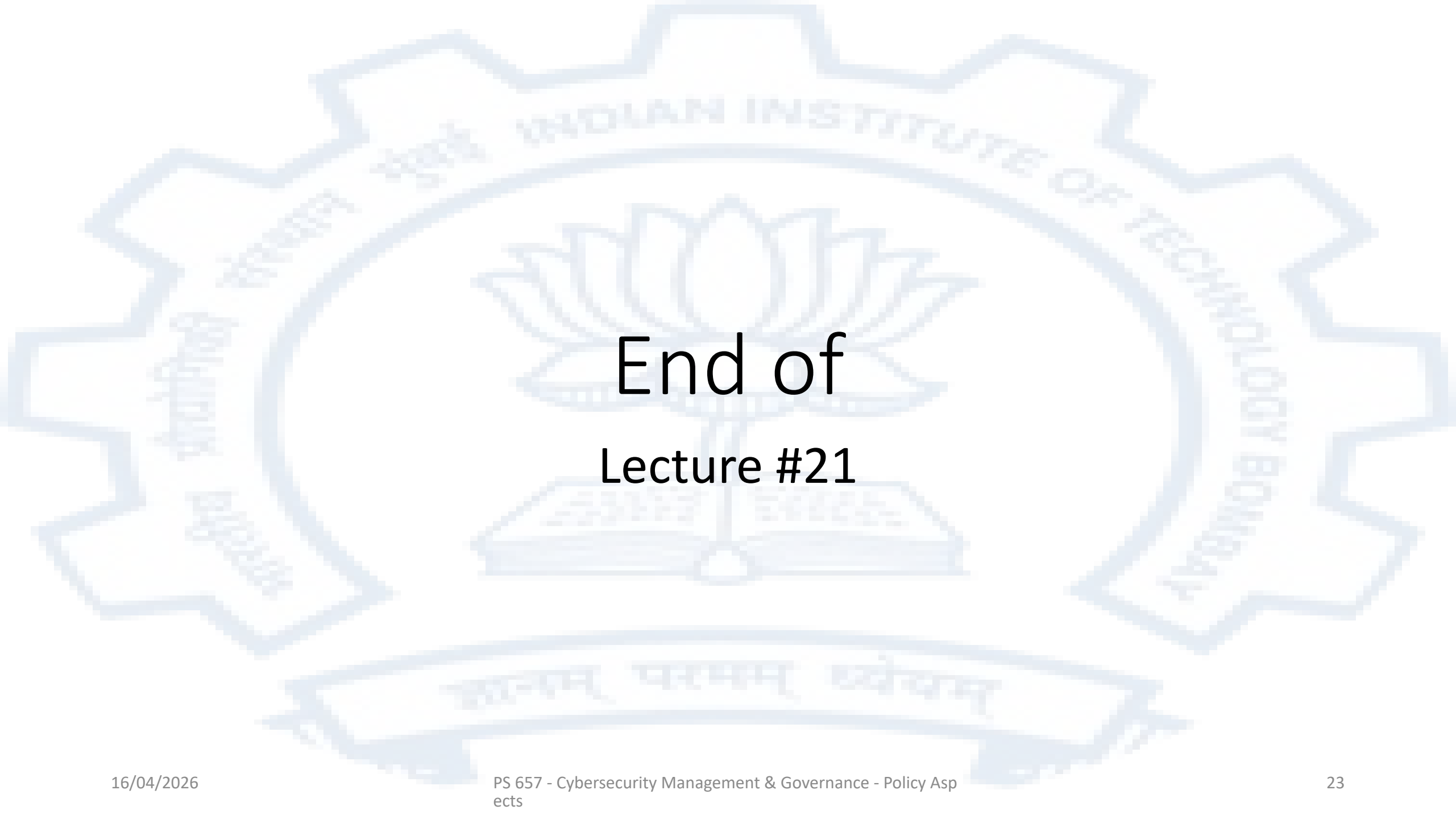
- IoT – Computers integrated with the real world
 - Cars
 - Aircraft
 - Plants
 - Medical equipment and implants
 - All of this is run with software and software has bugs and vulnerabilities. Software needs frequent updates. Authenticating updates is a critical challenge. Securing communication channels and operational security are also huge challenges
 - IoT devices are shipped with default passwords and may not have adequate cryptographic capabilities
- Computers integrated into humans
 - The targets are now biological systems. This opens a whole new frontier on the concept of privacy

Some demonstrated attacks

- [IoT Goes Nuclear: Creating a ZigBee Chain Reaction | Eyal Ronen](#)
- <https://conosco.com/industry-insights/blog/iot-security-breaches-4-real-world-examples>

To summarize

- In the technological space we are rapidly automating things that were thought to require human intelligence and cognitive capabilities
- We are bringing computers into decision making processes that were once the exclusive preserve of humans
- We are integrating computing elements into biological systems at a physiological level
- We are integrating computing into human minds
 - Brain computer interfaces + AI
- Cybersecurity failures can now have much more extensive impacts



End of Lecture #21